

DTSA 5514

Modern AI Models for Vision and Multimodal Understanding

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	MLOps Engineer (66.2 / 100)	55.5 / 100
SPECIALIZATION Computer Vision		

55

CURRENT AVG SCORE
across 6 job roles

88

POTENTIAL SCORE
with recommended additions

+47

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

This course delves into the cutting-edge realm of generative models for images and videos, including GANs and Diffusion Models. It will teach about multimodal foundational models such as CLIP, as well as applications for text-to-image and text-to-video generation. The course also addresses the issue of DeepFakes. Through both practical exercises and theoretical discussion, students will explore the ethical considerations, privacy concerns, and future trends in computer vision.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
MLOps Engineer	66.2	96.3	+30.1	HIGH
AI Engineer	63.2	93.9	+30.7	HIGH
Applied Scientist	58.7	91.5	+32.7	HIGH
Machine Learning Engineer	54.4	80.2	+25.8	MEDIUM
ML Engineer	50.5	78.8	+28.3	MEDIUM
Data Scientist	39.8	87.2	+47.4	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 5 of 6 roles

systems

Business Communication 1 of 6 roles

insights

Deep Learning 1 of 6 roles

cnn

Experiment Tracking 1 of 6 roles

tracking

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **HIGH** **Add System Design coverage** — required vocabulary in 5 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 **LOW** **Optional: Business Communication** — would broaden appeal to one additional role. Candidate terms: insights.

3 **LOW** **Optional: Deep Learning** — would broaden appeal to one additional role. Candidate terms: cnn.

4 **LOW** **Optional: Experiment Tracking** — would broaden appeal to one additional role. Candidate terms: tracking.

GAP KEYWORDS BY ROLE

MLOps Engineer

66.2 → 96.3 (+30.1 pts)

detection data model text systems learning machine tracking

AI Engineer

63.2 → 93.9 (+30.7 pts)

data learning build model systems problems technology detection

Applied Scientist

58.7 → 91.5 (+32.7 pts)

computational research literature problems data learning systems industry

Machine Learning Engineer

54.4 → 80.2 (+25.8 pts)

data learning tools science model machine systems information

ML Engineer

50.5 → 78.8 (+28.3 pts)

cnn media data learning machine systems model world

Data Scientist

39.8 → 87.2 (+47.4 pts)

data insights social build problems approaches

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.